

## L1.2 — Hardware: What You Need

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**Level:** Foundation (Free) · **Time:** 15 minutes · **Type:** Information — budget decision, no installation **Author:** Adi Ciobanu · LeYa Academy

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#### [Ideal] What you'll know by the end

- The minimum and recommended specs for your local AI
  - 3 budget options — from \$0 to \$200+
  - Why an external SSD is the safest choice
  - The complete checklist + how to back up properly
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#### [Content] Content

##### 1. Minimum vs recommended specs

|                   | Minimum               | Recommended                         |
|-------------------|-----------------------|-------------------------------------|
| <b>RAM</b>        | 8 GB                  | 16 GB                               |
| <b>Free space</b> | 50 GB                 | 100+ GB                             |
| <b>Processor</b>  | Any laptop from 2020+ | Intel i5 / AMD Ryzen 5 (recent gen) |
| <b>Storage</b>    | HDD works (but slow)  | NVMe SSD (much faster)              |

**Why does RAM matter most?** Because AI models load into memory. With 8 GB you can run small models (2-3 billion parameters). With 16 GB you have room for bigger models and more simultaneous processes. RAM is the one thing you can't "trick" — you either have enough or you don't.

Your current laptop is almost certainly enough. Any laptop bought in the last 5-7 years with 8 GB RAM or more will work.

*(Note: hardware links in this course may contain affiliate tags — you pay the same price, you support the project.)*

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##### 2. 3 budget levels

**Budget — \$0** Use the laptop you already have. Create a separate partition on the internal drive and install Linux there.

[Term] **Partitioning** = splitting your drive into two separate zones, like two rooms in one house. Windows stays in one, Linux installs in the other.

(!!) **Riskier:** if something goes wrong during partitioning, you could lose your Windows data. **Back up everything first.** Don't skip this. See section 5 for backup guidance.

**Mid-range — ~\$50 (*recommended*)** Buy a 256 GB external SSD and install Linux on it. Your laptop stays untouched. Windows boots normally when you unplug the SSD.

[Term] **External SSD** = a portable drive you connect via USB. Think of it as a big, fast USB stick. You don't modify anything on your laptop.

[x] **Safest.** Even if you accidentally wipe the SSD, your laptop is intact. This is the course recommendation.

**Recommended SSDs** (any 256GB+ model from these brands works): - Samsung T7 500GB (~\$50) — fast, reliable, compact - SanDisk Extreme 500GB (~\$45) — shock-resistant, good for travel - Kingston XS2000 500GB (~\$55) — very fast, USB-C - *Or any 256GB+ external SSD from Samsung, SanDisk, Kingston, Crucial, WD — ~\$25-50*

**Comfortable — \$200+** Buy a dedicated mini-PC that becomes your personal AI server. It runs 24/7 without tying up your laptop.

**Recommended mini-PCs:** - Beelink S12 Pro (~\$180) — Intel N100, 16GB RAM, solid entry-level - Minisforum UM350 (~\$220) — Ryzen 5, more processing power - *Or any mini-PC with 16GB RAM — make sure it has a USB port for installation*

[Ideal] **Ideal long-term.** After Level 1-3, when you want a system that runs non-stop, a mini-PC is the right investment. But you don't need it now.

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### 3. Why an external SSD?

Three simple reasons:

1. **You don't touch your existing system.** Windows stays exactly as it is. Linux lives on the external SSD.
2. **Natural dual-boot.** When the SSD is plugged in, you boot Linux. When it's not, you boot Windows. Simple.
3. **If something goes wrong, you unplug the SSD.** Zero consequences. Your laptop is untouched.

**How dual-boot works:** Your laptop's BIOS checks what devices are connected. If the external SSD is plugged in and has an operating system, it boots from there. If not, it boots Windows from the internal drive. You control this from the BIOS — you'll learn how in L1.3.

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### 4. Checklist before Lesson 3

Before you start installing Linux, make sure you have:

- ☐ **External SSD available** (*recommended: Samsung T7, SanDisk Extreme, Kingston XS2000 — any 256GB+ works*)
  - ☐ **USB stick of 8 GB or more** — this will become your Linux installation stick
- [Term] **Bootable USB stick** = a regular USB stick with Linux written to it in a special format, so your laptop can start from it. Any 8GB+ stick works.
- ☐ **Working laptop with a USB port**
  - ☐ (*Optional*) Ethernet cable — WiFi usually works, but some laptops need a cable for the first install

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### 5. (!! ) Before anything: back up

If you're going with the internal partition option (budget), backing up is not optional. It's mandatory.

**How to back up on Windows:** 1. Plug in an external HDD or open Google Drive (free — 15GB) 2. Copy these folders: Documents, Desktop, Pictures, Downloads (whatever you need) 3. Verify the files copied — open one on the external HDD before continuing

**No backup destination?** Don't use the partition method (budget). Wait until you have an external SSD. The risk is too high.

**On Windows 7 or 8?** It might work, but it's not fully tested. In L1.3 you'll need Rufus 3.22 (not the latest version, which requires Windows 10+). If you run into issues → reach out to support.

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(!!) **If you checked NO to any checklist item:** don't move on to L1.3. Fix it first. Without an external SSD (or a backup) and a USB stick, installation is impossible. You need ~\$30-50 for the minimum equipment.

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### [Exercise] Exercise

1. Check off the checklist above — do you have everything?
  2. Decide which budget option: \$0 / \$50 / \$200+. There's no wrong answer.
  3. If you don't have an external SSD and want to buy one, pick from the recommendations above.
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### [Checkpoint] Checkpoint — don't move on until you can answer

#### 1. What's the difference between minimum and recommended specs?

*Expected:* Minimum: 8 GB RAM, 50 GB free, any recent laptop — it works. Recommended: 16 GB RAM, NVMe SSD — faster, room for bigger models.

#### 2. Why is it safer to install Linux on an external SSD?

*Expected:* Because you don't touch the internal drive. If something goes wrong, you unplug the SSD and your laptop boots Windows normally. Zero risk.

#### 3. Do you have everything you need for Lesson 3? If not, what's missing?

*Expected:* Check the list. SSD + USB stick + laptop — all 3 checked? If not, what specifically is missing and when will you get it?

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(!!) **If you answered NO to question 3:** stop here. Re-read sections 2 (budgets) and 5 (backup). Without an external SSD + USB stick, Lesson 3 can't be done. You need ~\$30-50 for equipment. Come back when you have it.

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**All good? Move on to L1.3 — Linux Mint on an External SSD.**

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